MILITARY SPECIFICATION

PROPELLANT, SOLVENTLESS, TYPE M37 FOR PROJECTILE, 155MM, HE, ROCKET ASSISTED, M549 AND M549A1

This specification is approved for use by the U. S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication and packing of the M37 propellant for use in the solventless propellant grain.

1.2 Classification. The propellant shall be furnished in the following classes as specified:

Class 1 - Carpet Roll
Class 2 - Billet

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-181 - Wax (Candelilla)
MIL-N-244 - Nitrocellulose (For use in explosive)
MIL-N-246 - Nitroglycerine
MIL-T-301 - Triacetin (Glyceryl Triacetate)
MIL-N-3399 - 2-Nitrodiphenylamine
MIL-L-17700 - Lead Salicylate
MIL-L-17699 - Lead 2-Ethylhexoate
MIL-A-48078 - Ammunition Standards Quality Assurance Provisions, General Specification For

FSC 1376

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:24Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-286 - Propellants, Solids Sampling Examinations and Testing.

2.1.2 Other government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-199

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. The materials used in the manufacture of the propellant, shall conform to the requirements of the specifications listed in 2.1.

3.2 Composition. The composition of the propellant on a moisture free basis, shall consist of the constituent materials listed in Table I. A maximum of 35 percent rework propellant may be uniformly dispersed in each batch. However, where used, each batch must contain the same amount of rework material. Material is considered rework only if it has been previously extruded and may consist of clippings, ends, mill chips and whole or sectioned propellant grains which have not been wrapped and bonded with inhibiting material. The composition shall either be made into, Class 1, Carpet Roll, or Class 2, Billet, as specified in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:24Z
Check the source to verify that this is the current version before use.

TABLE I

PROPELLANT COMPOSITION

Material		Percent by weight
Nitrocellulose Grade A Type 1 made with Cellulose Cotton Linters (MIL-C-206)		50.0 Minimum
Nitroglycerine, Type F	MIL-N-246	36.2 Nominal
2-Nitrodiphenylamine	MIL-N-399	1.0 Minimum
Triacetin	MIL-T-301	9.7 Nominal
Lead Salicylate	MIL-L-17700	1.5 Nominal
Lead 2-Ethylhexoate	MIL-L-17699	1.5 Nominal
Wax, Candelilla	MIL-W-181	0.1 Nominal

3.3 Strand burning rate. The strand burning rate shall comply to Table II. In addition, strand burning rate shall be tested at +70°F for info only.

TABLE II

STRAND BURNING RATE

Pressure Pounds per Square Inch	Burning Rate Inches per Second	
	Lower Limit @ Minus 40 F	Upper Limit @ Plus 104 F
600	.359	.421
800	.414	.464
1000	.433	.450
1200	.368	.462
1400		.479
1600		.581
1800		.615
2000		.663

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:24Z
Check the source to verify that this is the current version before use.

3.4 Heat of explosion. The heat of explosion shall be no less than 940 cal/gm nor more than 970 cal/gm when determined as specified in 4.5.1.

3.5 Taliani test. The rate of evolution of gas from samples subjected to the test specified in 4.4.1.5 and 4.5.1 shall be such that the characteristics of a curve showing rate of evolution (mm of mercury versus time) meets the following:

a. Slope of curve at 100mm of mercury shall be less than 1.00.
b. Slope of the curve at 100 minutes shall be less than 0.90.
c. Time of 100mm of mercury shall be more than 100 minutes.

3.6 Moisture content. The moisture content shall be no greater than 0.50 percent when tested as specified in paragraph 4.4.1.4 and paragraph 4.5.1.

3.7 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. Quality Conformance Inspection shall be conducted on this item.

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4 Examination and testing. See MIL-A-48078.

4.4.1 Propellant testing.

4.4.1.1 Sampling. Each lot of propellant submitted for acceptance shall be sampled by the contractor under the direction of the Government inspector. Twenty five individual even-speed-rolled sheets shall be selected at random from each inspection lot (see 6.3). A sample at least 11 inches long by 4 inches wide shall be cut at random from each of the 25 sheets selected. The samples

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:24Z
Check the source to verify that this is the current version before use.

shall be numbered consecutively to facilitate proper testing sequence. From each of the samples above, test samples shall be selected and prepared for laboratory tests described in 4.4.1.2 through 4.4.1.5.

4.4.1.2 Test sample A - Heat of explosion determination.- Eight (8) samples shall be selected at random from the 25 samples designated. From each of the eight samples, a portion weighing at least 20.0 grams shall be reduced in size as described in Method 509.3 of MIL-STD-286 to pass through a U.S. No. 20 mesh screen. Each of the eight resulting samples shall be stored in a separate moisture-tight container.

4.4.1.3 Test sample B - Strand burning rate determination. From each of the sections after removal of specimens for test sample A, as many 1/8 to 3/16 inch wide strands as are needed to obtain one satisfactory strand from each sample shall be cut. One 1/8 to 3/16 inch wide strand shall be removed from the remaining 17 samples not utilized in 4.4.1.2. The remainder of each of the sections plus unsatisfactory strands (if any) shall be saved for test sample D. The twenty five 1/8 inch wide strands, from all 25 samples thus obtained, shall be further cut to a length as described in Method 803.1 of MIL-STD-286 and the remaining short lengths shall be retained for test sample C. The resulting strands shall be stored in moisture-tight containers and properly identified.

4.4.1.4 Test sample C - Moisture determination. Each of the short lengths of 1/8 inch wide strips remaining after the preparation of test sample B shall be cut to a length of 3.0 inches. Then each of these lengths shall be further reduced in size by cutting into approximately 1/8 inch squares. The resulting material (at least 20.0 grams in weight) shall be thoroughly blended, stored in a moisture-tight container and properly identified.

4.4.1.5 Test sample D - General chemical analysis and Taliani test. From the material remaining after the preparation of test sample B and test sample C, a section weighing about 4.0 grams shall be selected from each of the 25 samples. These sections shall be reduced in size as described in Method 509.3, MIL-STD-286, and blended thoroughly to give a uniform sample weighing at least 160 grams.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:24Z
Check the source to verify that this is the current version before use.

4.4.1.6 Rejection. Failure of propellant to meet any requirement of this specification shall be cause for rejection of the lot represented.

4.5 Methods of inspection. PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Chemical analysis. To determine conformance of the propellant with 3.2, a chemical analysis shall be carried out on the sample taken in 4.4.1.5 and the methods shall be as described in Table III.

TABLE III

TEST METHODS (MIL-STD-286)

Material	Method Number
Nitrocellulose	209.2
Nitroglycerine	208.1 or 208.4 or 208.5
2-Nitrodiphenylamine	218.4 or 208.4 or 208.5
Candelilla wax	228.1
Triacetin	222.1 or 208.4
Lead salicylate	225.1
Lead 2-Ethylhexoate	311.5
Moisture	101.5 or 103.5 or 103.1
Taliani Stability	406.1
Heat of explosion	802.1

4.5.2 Strand-burning rate. The strand-burning rate shall be determined in accordance with MIL-STD-286, Method 803.1 with the 25 samples taken in 4.4.1.3. The strands shall be tested in nitrogen at -40°F and at 140°F.

5. Packaging

5.1 Specific Requirements

5.1.1 LEVEL A - NOT APPLICABLE

5.1.2 LEVEL B - The propellant shall be packed in a suitable container, constructed, and labeled in accordance with applicable regulations of the Department of Transportation. The container shall be acceptable for transportation by common carrier. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:24Z
Check the source to verify that this is the current version before use.

shipping container and manner of packing shall be indicated in the contract or purchase order. The container shall be appropriately sealed to provide waterproof and moisture resistant protection to the item.

5.2 MARKING. In addition to any special markings required by the contract or purchase order, all markings shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The rocket propellant covered by this specification is intended for use in the 155MM Rocket Assisted Projectile (M549) propellant grain.

6.2 Procurement requirements. Procurement requirements of MIL-A-48078 shall apply except for First Article Sample.

6.2.1 Contract data requirements. In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all test data and one copy of the Propellant Description sheet shall be forwarded to: Commander, US Army Armament Munitions and Chemical Command, ATTN: DRSMC-QAR-Q (D), and one copy to: ATTN: DRSMC-LCU-CP (D), Dover, New Jersey 07801.

6.3 Surveillance test. A five pound sample of propellant from each carpet roll lot shall be forwarded to: U. S. Army Armament Munitions and Chemical Command, ATTN: DRSMC-LCE-P (D), Dover, New Jersey 07801. All samples shall be packed in air tight containers and shall be marked to show the propellant disignation, lot number,manufacturer, date of sampling, contract number, and number of pounds in the lot.

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1376-A091)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:24Z
Check the source to verify that this is the current version before use.